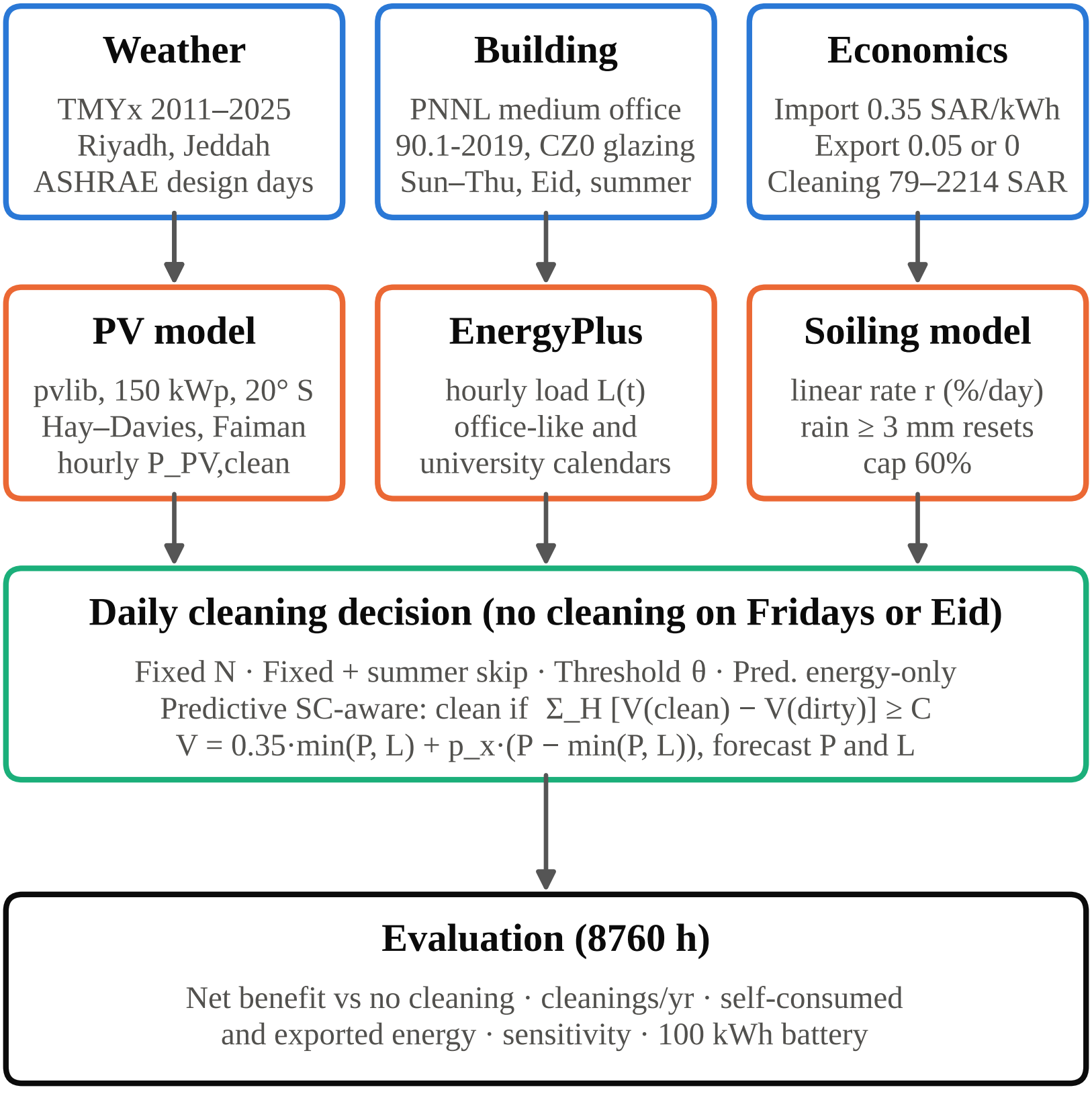
Weather
TMYx 2011–2025
Riyadh, Jeddah
ASHRAE design days
Building
PNNL medium office
90.1-2019, CZ0 glazing
Sun–Thu, Eid, summer
Economics
Import 0.35 SAR/kWh
Export 0.05 or 0
Cleaning 79–2214 SAR
PV model
pvlib, 150 kWp, 20° S
Hay–Davies, Faiman
hourly P_PV,clean
EnergyPlus
hourly load L(t)
office-like and
university calendars
Soiling model
linear rate r (%/day)
rain ≥ 3 mm resets
cap 60%
Daily cleaning decision (no cleaning on Fridays or Eid)
Fixed N · Fixed + summer skip · Threshold θ · Pred. energy-only
Predictive SC-aware: clean if Σ_H [V(clean) − V(dirty)] ≥ C
V = 0.35·min(P, L) + p_x·(P − min(P, L)), forecast P and L
Evaluation (8760 h)
Net benefit vs no cleaning · cleanings/yr · self-consumed
and exported energy · sensitivity · 100 kWh battery